

Module 1 - AI Fundamentals & AI Literacy

1.1 Introduction to AI

1.2 Generative AI and LLM fundamentals

1.3 Limitations of LLMs and Risks

1.4 Prompting fundamentals

1.5 Introduction to AI agents and the agentic stack

Module 2 - RAG implementation

2.1 RAG fundamentals

2.2 Basics of embeddings and semantic search

2.3 Vector databases

2.4 The framework landscape — LangChain & LlamaIndex

2.5 Document loading and chunking strategies

2.6 Embedding and indexing pipeline

2.7 Retrieval strategies

2.8 Prompt construction and generation

2.9 Building a basic RAG pipeline end-to-end

Module 3 - AI agents in LangChain

3.1 What is an AI agent — moving beyond RAG

3.2 Function calling and tool use

3.3 LangChain core abstractions for agents

3.4 Building and registering custom tools

3.5 Agent types and reasoning loops (ReAct, tool-calling)

3.6 Agent memory basics

3.7 Error handling, retries, and reliability

3.8 Building a basic agent end-to-end

Module 4 - Introduction to LangGraph

4.1 Why LangGraph — limitations of chain-based agents

4.2 LangGraph core concepts — nodes, edges, state

4.3 Building your first graph

4.4 Conditional edges and branching logic

4.5 Human-in-the-loop workflows

4.6 Multi-agent graphs

4.7 Building a LangGraph agent end-to-end

Module 5 - Tools, MCP and Memory in Agents

5.1 Why memory matters — statelessness problem

5.2 Short-term memory — conversation history and context window management

5.3 Long-term memory — vector store backed memory

5.4 Episodic vs semantic vs procedural memory types

5.5 Memory in LangGraph — state persistence across runs

5.6 MCP server and tools connections to Agents

5.7 Building an agent with persistent memory, tools and MCP

Module 6 - LLM and RAG evaluation

6.1 Why evaluation matters — moving beyond vibes

6.2 LLM evaluation fundamentals — metrics, golden datasets, LLM-as-judge

6.3 RAG-specific evaluation — retrieval quality (precision, recall, MRR)

6.4 Generation quality — faithfulness, relevance, groundedness

6.5 Building an evaluation pipeline

Module 7 - Advanced/production RAG

7.1 Failure modes in naive RAG

7.2 Advanced retrieval strategies

7.3 Query transformation and routing

7.4 LlamaIndex for production RAG

7.5 Observability, tracing, and monitoring

7.6 Cost optimization and latency management

7.7 Production RAG architecture patterns

Module 8 - Practical tasks

8.1 Build AI agent to communicate with your database - 'Talk to Data' - Use Athena + Bedrock Knowledge Base

8.2 Automate an entire ETL flow with DBT+Snowflake using AI agents - 'Agentic ETL Pipeline'

8.3 Automate monitoring, debugging and bug fix in an ETL pipeline using AI agents - 'Agentic Remediation Pipeline'